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Signed-Distance Fuzzy Logic Controller Adaptation Mechanism based MRAS Observer for Direct-Drive PMSG Wind Turbines Sensorless Control

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Abstract — **In this paper, a signed-distance fuzzy logic controller (*SDFLC*) adaptation mechanism is proposed to improve the permanent-magnet synchronous generator (*PMSG*) speed estimation performance of a model reference adaptive speed (*MRAS*) observer. In the conventional *MRAS* observer, a constant gain *PI* controller is used to drive the output error vector between the reference and the adaptive model to zero. One of the improvement noted in literature is the two-input fuzzy logic controller (*FLC*), this latter have greatly improved the speed estimation performance of the *MRAS* observer. The major drawback of the two-input fuzzy logic controller is the high fuzzy rules number, the thing that increase significantly the tuning and the control complexity. In the proposed adaptation mechanism, the two-input *FLC* are converted into single input named *signed distance.* This variable contains knowledge of all process state variables of the conventional *FLC*, and instead of creating and designing the fuzzy rules using a two-dimensional space of the phase plane, the new fuzzy rules will be designed only in one-dimensional rule table, this method allows greatly reducing the number of fuzzy rules and easily tuning the controller. A detailed comparison between the *PI*, *FLC* and *SDFLC MRA*S observer in open loop mode has been performed under step wind speed variation. As a second test, the *MRAS* observer based on *SDFLC* adaptation mechanism has been used in a vector super-twisting algorithm (*STA*) control strategy for a direct-drive *PMSG* wind turbines.**

Keywords: *MRAS Observer, Sensorless Control, Fuzzy Logic Controller, Wind Turbine, PMSG,MPPT.*

I. INTRODUCTION

Sensorless control is one of the most important issues in wind energy conversion systems (*WECS*), reducing the control scheme cost and increasing the system reliability is the main objective. Many drawbacks are associated with the use of mechanical sensors such as rotor position and rotational speed, these later increases the failure rate of a *WECS* [1] and result in additional costs for repairing the failed components. The best solution to avoid these problems is to use an estimator or an observer [2].

In literature, several rotational speed observers have been developed for a permanent-magnet synchronous generator (*PMSG*). In [2], a comparative study between: synchronous reference frame (*SRF*) phase locked loop, speed estimation algorithm starting from the measured current and voltage of the boost rectifier, extended kalman filter and linear kalman filter has been made for a *PMSG* controlled by a boost rectifier working in discontinuous conduction mode. A comparative study between a sliding mode, artificial neural network and model reference adaptive speed observers has been presented in [3]. In [4], a nonlinear Luenberger-like observer was designed to estimate the mechanical variables. An extended *EMF* estimator is used in sensorless control in [5]. A hybrid observer combining a higher-order sliding mode observer (*HOSM*) based on modified super twisting algorithm (*STA*) and a rotor flux estimation and using a switching algorithm in order to select the appropriate observer is proposed in [6]. In [7], a speed estimator based on Angle Tracking Observer (*ATO*) is developed for *PMSG* sensorless control.

The model reference adaptive speed (*MRAS*) observer is the most attractive technique because of its simplicity and its high estimation performance [8][9]. In an *MRAS* observer, a reference model and an adaptive model are connected in parallel; the output of the adaptive model should converge to the reference model output using an appropriate adaptation mechanism. The adaptive mechanism is the most essential element. The classical *MRAS* observer is based on constant gain proportional integral controller. However, it has been shown in literature that the technique based on *PI* controller cannot provide a high estimation performance [10]. Many adaptation mechanism improvements have been discussed in literature. In [11], an improved *MRAS* observer based on sliding mode controller has been proposed for a permanent magnet synchronous generator. In [12], two novel nonlinear adaptation mechanisms have been proposed instead of the conventional *PI* controller for estimating the rotational speed of an induction motor (*IM*), the first one is based on fuzzy logic controller and the second one is based on sliding mode controller. In [13], a type-1 fuzzy logic controller (*TIFLC*) and type-2 fuzzy sets (*T2FLC*) were implemented for sensorless induction motor control based *MRAS* observer. A linear artificial neural networks (*ANN*) adaptation mechanism has been proposed in [14] for induction motor sensorless drive.

In this paper, a signed-distance fuzzy logic controller (*SDFLC*) is proposed as an adaptation mechanism for the *MRAS* observer. The idea here consists of converting the two-input of the conventional fuzzy logic controller (*CFLC*) into single input. Comparing to the conventional two-input fuzzy logic controller adaptation mechanism, the proposed technique allows greatly reducing the number of fuzzy rules and an easiest tuning. By using the *SDFLC,* the fuzzy rules can be designed and created only using a one-dimensional rule table instead of two-dimensional space of the phase plane. According to [15], the signed distance (single input) fuzzy logic controller is equivalent to the pseudo sliding mode

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controller (*SMC*) without the equivalent control input. This means that, the signed distance fuzzy logic controller requires less information comparing to the sliding mode family controllers. The theoretical study has shown that instead of using *49* fuzzy rules the new *MRAS* observer can be designed only using seven fuzzy rules. The performance of the *PI*, *CFLC* and *SDFLC MRAS* observer were tested firstly in open loop mode under random step wind profile. As a second test, the proposed observer has been integrated in a vector super-twisting algorithm (*STA*) control strategy in order to avoid the use of a mechanical speed sensor. The overall sensorless control scheme has been applied on the wind electric water pumping system based on *PMSG* and permanent-magnet *DC* motor shown in Figure (1). Computer simulation using *Matlab/Simulink software* confirm the effectiveness of the proposed *MRAS* observer in open and closed loop mode.

II. System Description and Modeling

The wind turbine, the permanent magnet synchronous generator, the permanent magnet *DC* motor and the centrifugal pump models are described by the following equations [1][3][5]:

$$p_v = \frac{(\rho.A.{v_w}^3)}{2} \quad (1)$$

$$p_m = \frac{1}{2}C_p(\lambda,\beta).\rho.A.{v_w}^3 \quad (2)$$

$$\lambda = \frac{\Omega_m^G R}{v_w} \quad (3)$$

$$\begin{cases} C_p(\lambda,\beta) = C_1\left(\frac{C_2}{\gamma} - C_3.\beta - C_4\right)e^{\frac{-C_5}{\gamma}} + C_6.\lambda \\ \frac{1}{\gamma} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3+1} \end{cases} \quad (4)$$

$$T_m^G = \frac{p_m}{\Omega_m^G} = \frac{1}{2.\Omega_m^G}C_p(\lambda,\beta).\rho.A.{v_w}^3 \quad (5)$$

$$\begin{cases} T_m^G = J^G\dot{\Omega}_m^G + f.\Omega_m^G + T_{em}^G \\ J^G = J_{turbine}^G + J_g^G \end{cases} \quad (6)$$

$$\begin{cases} V_{ds} = R_sI_{ds} + L_d\dot{I}_{ds} - \omega_r\psi_{qs} \\ V_{qs} = R_sI_{qs} + L_q\dot{I}_{qs} + \omega_r\psi_{ds} \\ \psi_{ds} = L_dI_{ds} + \psi_0 \\ \psi_{qs} = L_qI_{qs} \end{cases} \quad (7)$$

$$\begin{cases} L_d\dot{I}_{ds} = V_{ds} - R_sI_{ds} + \omega_rL_qI_{qs} \\ L_q\dot{I}_{qs} = V_{qs} - R_sI_{qs} - \omega_rL_dI_{ds} - \psi_0\omega_r \end{cases} \quad (8)$$

$$T_{em}^G = \frac{3}{2}p\left[(L_d - L_q)I_{ds}I_{qs} + \psi_0I_{qs}\right] \quad (9)$$

$$T_{em}^G = \frac{3}{2}p\psi_0I_{qs},\ (L_d = L_q) \quad (10)$$

$$\begin{cases} V_{aM} = R_{aM}I_{aM} + L_{aM}\dot{I}_{aM} + e \\ T_e^M = K_tI_{aM} \\ e = K_e\Omega_m^M \\ T_e^M - T_L^M = J^M\dot{\Omega}_m^M + B_m.\Omega_m^M + T_f^M \end{cases} \quad (11)$$

$$T_L^M = a.{\Omega_m^M}^n + b \quad (12)$$

Where p_v is the wind power, ρ is the air density, A is the circular area, v_w is the wind speed, p_m is the mechanical power, C_p is the power coefficient, β is the pitch angle, λ is the tip speed ratio, Ω_m^G is the turbine rotor speed, R is the turbine radius, T_m^G is the mechanical torque, T_{em}^G is the electromagnetic torque produced by the generator, f is the friction coefficient, J^G is the total moment of inertia of the rotating parts, L_d, L_q are the inductances of the generator on the q and d axis, R_s is the stator resistance, ψ_0 is the permanent magnetic flux, ω_r is the the electrical rotating speed of the *PMSG* which is given by $\omega_r = p.\Omega_m^G$, p is the number of pole pairs, R_{aM} is the armature winding resistance, L_{aM} is the armature self-inductance, I_{aM} is the motor armature current, V_{aM} is the applied voltage, e is the back *e.m.f* of the *PMDC* motor, K_e is the voltage constant, Ω_m^M is the angular speed, K_t is the torque constant, J^M is the moment of inertia, B_m is the viscous torque constant, T_f^M is the torque constant for rotational losses, T_e^M, T_L^M are the electromagnetic torque and load torque respectively, and a, b are the constants of the pump.

Figure 1. Studied Wind Electric Water Pumping System

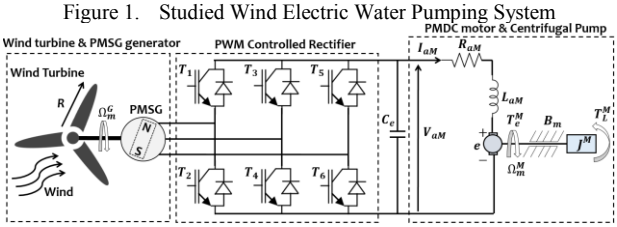
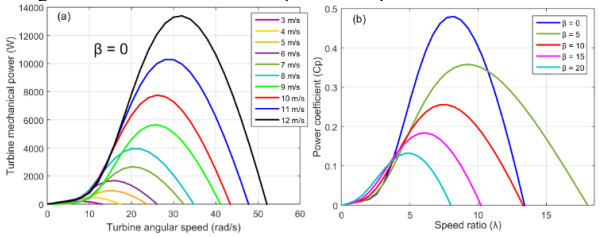


Figure 2. Tubine mechanical power and power cofficient curves



III. Model Reference Adaptive Speed (*MRAS*) Observer

A. PI MRAS Speed Observer

The model reference adaptive speed observer of a *PMSG* as shown in Figure (3) is composed of two models, a reference model represented by the equations (13) and (14), and an adaptive model represented by the equations (15) and (16).

$$\psi_{ds} = L_dI_{ds} + \psi_0 \quad (13)$$

$$\psi_{qs} = L_qI_{qs} \quad (14)$$

$$\hat{\psi}_{ds} = \int\left(V_{ds} - R_s\hat{I}_{ds} + \hat{\omega}_rL_q\hat{I}_{qs}\right)dt + \psi_0 \quad (15)$$

$$\hat{\psi}_{qs} = \int\left(V_{qs} - R_s\hat{I}_{qs} - \hat{\omega}_rL_d\hat{I}_{ds} - \psi_0\hat{\omega}_r\right)dt \quad (16)$$

The error (ε_{ω_r}) between the reference and adaptive model outputs defined by the equation (17) is used as input to an adaptive mechanism. In the conventional *MRAS* observer, a constant gain proportional integral controller is used as an adaptation mechanism [3][13]. The objective is to drive the error (ε_{ω_r}) to zero, so that the estimated rotational speed $\hat{\omega}_r$ converge to its original value.

$$\varepsilon_{\omega_r} = \hat{\psi}_{ds}\psi_{qs} - \hat{\psi}_{qs}\psi_{ds} \quad (17)$$

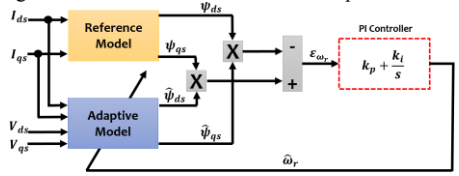
The error in the *d-q* coordinates is defined as: $[\varepsilon_{\omega_{r1}}, \varepsilon_{\omega_{r2}}]^T = [\hat{\psi}_{ds} - \psi_{ds}, \hat{\psi}_{qs} - \psi_{qs}]^T$. The estimated speed using a *PI* controller adaptation mechanism based *MRAS* observer is given as follows:

$$\hat{\omega}_r = k_p\left(\varepsilon_{\omega_{r1}}.\hat{\psi}_{qs} - \varepsilon_{\omega_{r2}}.\hat{\psi}_{ds}\right) + k_i\int\left(\varepsilon_{\omega_{r1}}.\hat{\psi}_{qs} - \varepsilon_{\omega_{r2}}.\hat{\psi}_{ds}\right)dt + \omega_r(0) \quad (18)$$

Where:

$$\varepsilon_{\omega_{r1}}.\hat{\psi}_{qs} - \varepsilon_{\omega_{r2}}.\hat{\psi}_{ds} = \psi_{qs}.\hat{\psi}_{ds} - \psi_{ds}.\hat{\psi}_{qs} \qquad (19)$$

Figure 3. Conventional *PMSG MRAS* speed observer



B. Two-input FLC MRAS Speed Observer

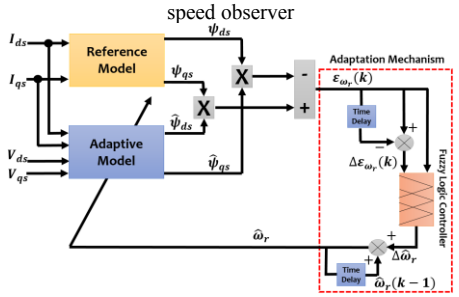
The fuzzy logic controller is one of the existed adaptation mechanism improvement of an *MRAS* observer (Figure (4)) [12][13]. This latter is used to overcome the conventional *MRAS* observer drawbacks. The structure of a fuzzy logic controller consists of *Fuzzification process, an Inference engine, a knowledge base and a Defuzzification process*. The conventional fuzzy logic controller (*FLC*) adaptation mechanism receives two input variables, the error between the adaptive and reference *d-q* axis stator flux model (ε_{ω_r}) (Equation (17)) and the change of the error ($\Delta\varepsilon_{\omega_r}$) which is calculated by the following equation:

$$\Delta\varepsilon_r = \varepsilon_{\omega_r}(k) - \varepsilon_{\omega_r}(k-1) \qquad (20)$$

The fuzzy logic controller output provides the change of the estimated *PMSG* rotational speed $\Delta\hat{\omega}_r$. The estimated generator rotational speed is obtained as follows:

$$\hat{\omega}_r = \hat{\omega}_r(k-1) + \Delta\hat{\omega}_r(k) \qquad (21)$$

Figure 4. Two-input *FLC* adaptation mechanism of *PMSG MRAS* speed observer



Figures (5),(6) and (7) illustrate the fuzzy logic membership functions for the input and output variables. Seven fuzzy subsets were used for input and output membership functions: *NB* (*Negative Big*), *NM* (*Negative Medium*), *NS* (*Negative Small*), *ZE* (*Zero*), *PS* (*Positive Small*), *PM* (*Positive Medium*) and *PB* (*Positive Big*). *49* fuzzy rules were designed for the used fuzzy logic controller (Table (1)). *Mamdani* method with *Max-Min* was used to carry out the fuzzy inference.

The designed fuzzy control rules for the two-input *FLC* have the form of "***If-Then***" with "***And***" logical operator, such as: ***if*** ε_{ω_r} *is NB* ***And*** $\Delta\varepsilon_{\omega_r}$ *is PM* ***Then*** $\Delta\hat{\omega}_r(k)$ *is PS.* The *Defuzzification* process is performed using the center of gravity method and it is given by the following expression:

$$y = \frac{\Sigma_{j=1}^{n} y_j \mu(y_j)}{\Sigma_{j=1}^{n} \mu(y_j)} \qquad (22)$$

Figure 5. The error ε_r membership functions

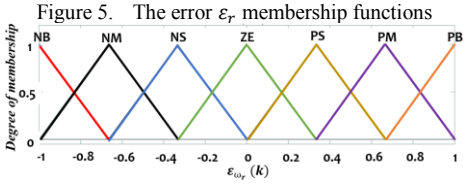


Figure 6. The error change $\Delta\varepsilon_r$ membership functions

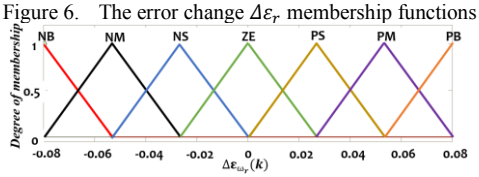


Figure 7. Change in the estimated speed $\Delta\hat{\omega}_r$ membership functions

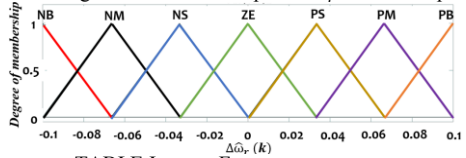


TABLE I. FUZZY RULES TABLE

$\Delta\varepsilon_{\omega_r}$ \ ε_{ω_r}	***NB***	***NM***	***NS***	***ZE***	***PS***	***PM***	***PB***
NB	*PB*	*PB*	*PM*	*PM*	*PS*	*PS*	*ZE*
NM	*PM*	*PM*	*PS*	*PS*	*PS*	*ZE*	*NS*
NS	*PM*	*PM*	*PS*	*PS*	*ZE*	*NS*	*NS*
ZE	*PM*	*PM*	*PS*	*ZE*	*NS*	*NM*	*NB*
PS	*PM*	*PS*	*ZE*	*NS*	*NS*	*NM*	*NM*
PM	*PS*	*ZE*	*NS*	*NS*	*NS*	*NM*	*NM*
PB	*ZE*	*NS*	*NS*	*NS*	*NM*	*NM*	*NB*

C. Signed distance FLC MRAS Speed Observer

The signed distance fuzzy logic controller (*SDFLC*) or also called single input fuzzy logic controller (*SIFLC*) [15] is the simplified version of the conventional two-input fuzzy logic controller presented in the previous section. The main idea here is about reducing the number of fuzzy control rules by considering only a single input variable instead of two input variables. In the previous part, the estimated rotational generator speed is obtained by using two fuzzy input variables, the first one is the output error vector ($\varepsilon_{\omega_r}(k)$) between the reference and the adaptive model, and the second one is the derivate (change) of the error ($\Delta\varepsilon_r(k)$). In the single input fuzzy logic controller, instead of the two fuzzy inputs ($\varepsilon_{\omega_r}, \dot{\varepsilon}_{\omega_r}$) a single input fuzzy variable called *signed distance* is introduced (Figure (8) [15].

Defining the switching line as:

$$S_1: \dot{\varepsilon}_{\omega_r} + \zeta\varepsilon_{\omega_r} = 0 \qquad (23)$$

The new introduced variable is named *signed distance*. Let $D(\varepsilon_{\omega_r}, \dot{\varepsilon}_{\omega_r})$ be the intersection point of the switching line and the line perpendicular to the switching line from an operating point $G(\varepsilon'_{\omega_r}, \dot{\varepsilon}'_{\omega_r})$, as illustrated in Figure (9). The distance between the points $D(\varepsilon_{\omega_r}, \dot{\varepsilon}_{\omega_r})$ and $G(\varepsilon'_{\omega_r}, \dot{\varepsilon}'_{\omega_r})$ can be calculated as follows:

$$d_1 = \left[(\varepsilon_{\omega_r} - \varepsilon'_{\omega_r})^2 + (\dot{\varepsilon}_{\omega_r} - \dot{\varepsilon}'_{\omega_r})^2\right]^{\frac{1}{2}} = \frac{\left|\dot{\varepsilon}'_{\omega_r} + \zeta\varepsilon'_{\omega_r}\right|}{\sqrt{1+\zeta^2}} \qquad (24)$$

The generalization of the equation (24) of d_1 is given as follows:

$$d = \frac{\left|\dot{\varepsilon}_{\omega_r} + \zeta\varepsilon_{\omega_r}\right|}{\sqrt{1+\zeta^2}} \qquad (25)$$

For a general point $G(\varepsilon_{\omega_r}, \dot{\varepsilon}_{\omega_r})$, the signed distance d_s can be rewritten as follows:

$$d_s = sign(S_1)\frac{\left|\dot{\varepsilon}_{\omega_r} + \zeta\varepsilon_{\omega_r}\right|}{\sqrt{1+\zeta^2}} = \frac{\dot{\varepsilon}_{\omega_r} + \zeta\varepsilon_{\omega_r}}{\sqrt{1+\zeta^2}} = \frac{S_1}{\sqrt{1+\zeta^2}} \qquad (26)$$

Where:

$$sign(S_1) = \begin{cases} 1 & for \quad S_1 > 0 \\ -1 & for \quad S_1 < 0 \end{cases} \qquad (27)$$

Since the sign of the control input is negative for $S_1 > 0$ and positive for $S_1 < 0$, and its magnitude is proportional to the distance from the line $S_1 = 0$, we can conclude that:

$$\Delta\hat{\omega}_r \propto d_s \qquad (28)$$

Hence, the control action can be determined by d_s only and the fuzzy rule table can be designed only using one-dimensional space instead of two-dimentional space like Table (1).

Figures (10) and (11) show respectively the signed distance d_s input fuzzy logic membership functions and the change in the estimated speed $\Delta\hat{\omega}_r$ output fuzzy logic membership functions. The seven used linguistic variables are: *NB* (*Negative Big*), *NM* (*Negative Medium*), *NS* (*Negative Small*), *ZE* (*Zero*), *PS* (*Positive Small*), *PM* (*Positive Medium*) and *PB* (*Positive Big*). The final estimated rotational speed is obtained using the equation (21). *Mamdani* method is utilized to carry out the fuzzy inference, and the defuzzification operation can be performed by using the center of gravity (or centroid method) to compute the fuzzy controller output (Equation (22)). Seven fuzzy rules were designed for the signed distance fuzzy logic controller adaptation mechanism, the fuzzy rules have the form of "***If-Then***", and are represented as below:

1. ***If*** d_s *is NB* ***Then*** $\Delta\hat{\omega}_r(k)$ *is PB*
2. ***If*** d_s *is NM* ***Then*** $\Delta\hat{\omega}_r(k)$ *is PM*
3. ***If*** d_s *is NS* ***Then*** $\Delta\hat{\omega}_r(k)$ *is PS*
4. ***If*** d_s *is ZE* ***Then*** $\Delta\hat{\omega}_r(k)$ *is ZE*
5. ***If*** d_s *is PS* ***Then*** $\Delta\hat{\omega}_r(k)$ *is NS*
6. ***If*** d_s *is PM* ***Then*** $\Delta\hat{\omega}_r(k)$ *is NM*
7. ***If*** d_s *is PB* ***Then*** $\Delta\hat{\omega}_r(k)$ *is NB*

The rules form for the signed distance fuzzy logic controller R_{new} and the two-input fuzzy logic controller R_{old} are as follow:

$$\begin{cases} R_{new}: \textbf{\textit{If}}\ d_s\ is\ NB\ \textbf{\textit{Then}}\ \Delta\hat{\omega}_r(k)\ is\ PB \\ R_{old}: \textbf{\textit{If}}\ \varepsilon_{\omega_r}\ is\ NB\ \textbf{\textit{And}}\ \Delta\varepsilon_{\omega_r}\ is\ PM\ \textbf{\textit{Then}}\ \Delta\hat{\omega}_r(k)\ is\ PS \end{cases}$$

By comparing the two-input *FLC* and the signed disctance *FLC,* we can notice that the total number of fuzzy rules is greatly reduced, from *49* fuzzy to *7* fuzzy rules. An easy fuzzy control rules tuning can be made to improve the speed estimation accuracy using the signed distance variable d_s.

Figure 8. Signed distance *FLC* adaptation mechanism based *MRAS* observer

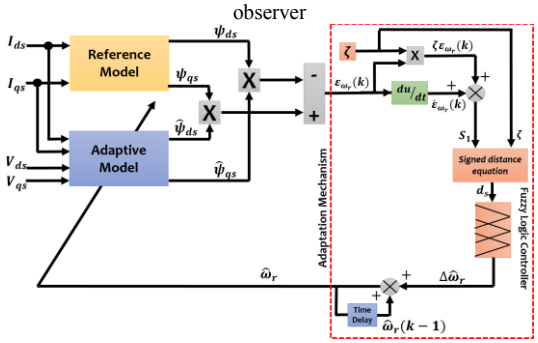


Figure 9. Derivation of a signed distance

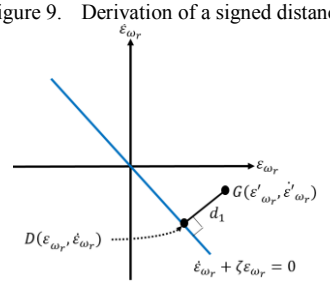


Figure 10. Signed distance (d_s) membership functions

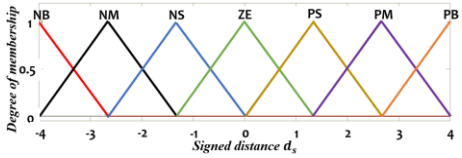
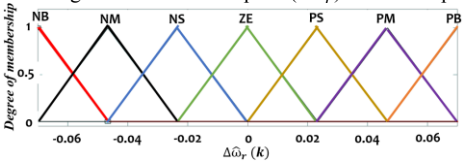


Figure 11. Change in the estimated speed ($\Delta\hat{\omega}_r$) membership functions



- *Stability analysis of the signed distance fuzzy logic controller*

Considering the *n*th order linear or nonlinear control system described by the following state equation:

$$z^{(n)} = g(z,t) + b(z,t)u(t) + d(t)$$
$$y = z \qquad (29)$$

With $z = [z_1, z_2, \dots, z_n]^T = [z, \dot{z}, \dots, z^{n-1}]^T$, where $g(z,t)$ and $b(z,t)$ are the nonlinear functions describing the system dynamics, $d(t)$ is the unknown external disturbance, $z(t) \in \Re^n$ is a state vector, $u(t) \in \Re$ and $y(t) \in \Re$ are respectively the system input and output. The control objective is to force $y(t)$ tracking a given bounded reference input signal z_d, which means for the *MRAS* observer minimizing the error between the estimated and reference *d-q* axis stator flux, in other words, driving the output error vector to zero. The tracking error as seen before is defined as follows:

$$\varepsilon_{\omega_r}(t) = z(t) - z_d(t) = \hat{\psi}_{ds}\psi_{qs} - \hat{\psi}_{qs}\psi_{ds}$$
$$= [\varepsilon_{\omega_r}(t), \dot{\varepsilon}_{\omega_r}(t), \dots, \varepsilon_{\omega_r}^{(n-1)}(t)]^T \qquad (30)$$

For the single input (signed distance) fuzzy logic controller extended version, the switching line S_1 can be expressed by the following hyperplane [15]:

$$S_1: \varepsilon_{\omega_r}{}^{(n-1)} + \zeta_{n-1}\,\varepsilon_{\omega_r}{}^{(n-2)} + \cdots + \zeta_2\dot{\varepsilon}_{\omega_r} + \zeta_1\varepsilon_{\omega_r} = 0 \qquad (31)$$

Similarly to the equation (30), the new general signed distance D_s is expressed as follows:

$$D_s = \frac{\varepsilon_{\omega_r}{}^{(n-1)} + \zeta_{n-1}\,\varepsilon_{\omega_r}{}^{(n-2)} + \cdots + \zeta_2\dot{\varepsilon}_{\omega_r} + \zeta_1\varepsilon_{\omega_r}}{\sqrt{1 + \zeta_{n-1}^2 + \cdots + \zeta_2^2 + \zeta_1^2}} \qquad (32)$$

In this case, D_s represents the signed distance from the operating point to the switching hyperplane.

According to [15], the control output generated by the one-dimensional fuzzy rules space of the single input fuzzy logic controller is linear. It can be observed from Figure (12) that also for the *SDFLC MRAS* observer fuzzy rules (from *1* to *7)* the interferences rules visualization is approximately linear, then it can be said that:

$$u = \Delta\hat{\omega}_r = -K_D D_s \qquad (33)$$

Where $K_D > 0$ is a positive constant obtained by the linear fuzzy logic controller.

Let assume G and D a known upper bounds for the unknown dynamics functions g and d, the control gain function b is positive and it has also a known bound.

The stability analysis of the signed distance (single input) fuzzy logic controller is carried out according and as shown in [15].

Proposition 1: If the following conditions are satisfied:

$$|g| \le G, |d| \le D, 0 < b_{min} \le b \le b_{max} \quad (34)$$

and the constant K_D satisfies the following inequality [15]:

$$K_D |D_s| \ge b_{min}^{-1} \left(G + D + z_d^{(n)} + \sum_{i=1}^{n-1} \zeta_i \varepsilon_{\omega_r}^{(i)} + \upsilon \right), \quad \upsilon \ge 0 \quad (35)$$

then the signed distance fuzzy logic controller is stable in the sense of *Lyapunov*.

Proof: The stability analysis is analyzed and verified by considering the following *Lyapunov* function:

$$V = \frac{1}{2} D_s^2 \quad (36)$$

The derivative of V gives:

$$\begin{aligned} \dot{V} &= D_s \dot{D}_s \\ &= \frac{D_s}{\sqrt{1 + \zeta_{n-1}^2 + \cdots + \zeta_2^2 + \zeta_1^2}} \left(\varepsilon_{\omega_r}^{(n)} + \sum_{i=1}^{n-1} \zeta_i \varepsilon_{\omega_r}^{(i)} \right) \\ &= \frac{D_s}{\sqrt{\vartheta}} \left(g + bu + d - z_d^{(n)} + \sum_{i=1}^{n-1} \zeta_i \varepsilon_{\omega_r}^{(i)} \right) \\ &= \frac{D_s}{\sqrt{\vartheta}} \left(g - bK_D D_s + d - z_d^{(n)} + \sum_{i=1}^{n-1} \zeta_i \varepsilon_{\omega_r}^{(i)} \right) \\ &= \frac{D_s}{\sqrt{\vartheta}} \left(g - bK_D |D_s| sgn(D_s) + d - z_d^{(n)} + \sum_{i=1}^{n-1} \zeta_i \varepsilon_{\omega_r}^{(i)} \right) \end{aligned} \quad (37)$$

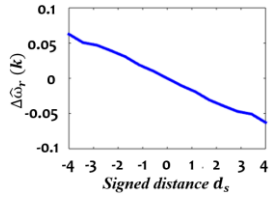
Where $\sqrt{\vartheta} = \sqrt{1 + \zeta_{n-1}^2 + \cdots + \zeta_2^2 + \zeta_1^2}$.

Substituting the Equation (35) into Equation (37), we obtain:

$$\dot{V} \le -\frac{\upsilon}{\sqrt{\vartheta}} |D_s| \quad (38)$$

We conclude that the signed distance (single input) fuzzy logic controller is stable in the sense of *Lyapunov*.

Figure 12. Visualization of the inference rules



IV. PMSG SIDE CONVERTER CONTROL

A vector control strategy is considered in this paper (Figure (13)). The main objective is controlling the generator speed, that can be done by adjusting the electromagnetic torque T_{em}^G to its reference $T_{em}^{G\,*}$. From the equation $I_{qs}{}^* = \frac{2}{3p\psi_0} T_{em}^{G\,*}$, we can notice that by acting on the q-axis stator current we can regulate the electromagnetic torque to its reference. Note that, the d-axis stator current is forced to zero to achieve a maximum generator torque [3]. In order to realize *MPPT*, the optimal (reference) rotational speed is calculated using $\Omega_{mopt}^G = {(\lambda^* . v_w)}/{R}$, where λ^* represents the optimum tip-speed ratio.

In such vector control strategy, we distinguish two control loops: an outer control loop used for controlling the generator speed Ω_m^G to its optimal reference value Ω_{mopt}^G and generating the q-axis reference stator current $I_{qs}{}^*$, and an inner control loop used to regulate the d-q axis stator current components (I_{qs}, I_{ds}) to their references $I_{ds}{}^*$ and $I_{qs}{}^*$ $(With: I_{ds}{}^* = 0)$. In the present control strategy, the controllers were designed using super-twisting algorithm (*STA*) . The *STA* is an exception version of high-order sliding mode controllers, its main advantage lies in the fact that it does not require any information on the time derivative of the sliding variable [17][18], while in the case of implementing the rth-order sliding mode controller, the *SMC* needs knowledge of $\sigma, \dot{\sigma}, \ddot{\sigma}, \ldots, \sigma^{(r-1)}$ (σ : sliding surface), the thing that result in a significant increasing information demand.

The sliding surface of the super-twisting algorithm speed controller is chosen as follows:

$$\sigma_{\Omega_m^G} = \Omega_{mopt}^G - \widehat{\Omega}_m^G \quad (39)$$

It follows that

$$\begin{cases} \dot{\sigma}_{\Omega_m^G} = \dot{\Omega}_{mopt}^G - \dot{\widehat{\Omega}}_m^G = \dot{\Omega}_{mopt}^G - \dfrac{T_m^G - f.\widehat{\Omega}_m^G - T_{em}^G}{J^G} \\ \ddot{\sigma}_{\Omega_m^G} = \varrho_1(t,x) + \Upsilon_1(t,x) \dot{T}_{em}^{G\,*} \end{cases} \quad (40)$$

Where $\widehat{\Omega}_m^G$ is the estimated rotational speed provided by the model reference adaptive speed observer detailed in the previous section, $\varrho_1(t,x)$ and $\Upsilon_1(t,x)$ represent the uncertain bounded functions that satisfy

$$\varrho_1(t,x) > 0,\ |\varrho_1(t,x)| > \Phi_1, 0 < \Gamma_{m1} < \Upsilon_1 < \Gamma_{M1}$$

The super-twisting algorithm control law contains two parts, a continuous function of the sliding surface $(\sigma_{\Omega_m^G})$ and the integral of a discontinuous control action [18]:

$$\begin{aligned} T_{em}^{G\,*} &= \upsilon_1 + \upsilon_2 \\ \begin{cases} \dot{\upsilon}_1 = -\alpha_1 sign(\sigma_{\Omega_m^G}) \\ \upsilon_2 = -\beta_1 \left| \sigma_{\Omega_m^G} \right|^{\rho} sign(\sigma_{\Omega_m^G}) \end{cases} \end{aligned} \quad (41)$$

Replacing $I_{qs}{}^* = \frac{2}{3p\psi_0} T_{em}^{G\,*}$ in equation (41), the super-twisting algorithm speed controller can be rewritten as follows:

$$\begin{aligned} I_{qs}{}^* = \frac{2}{3p\psi_0} (&-\beta_1 \left| \sigma_{\Omega_m^G} \right|^{\rho} sign\left(\sigma_{\Omega_m^G}\right) \\ &- \int \alpha_1 sign\left(\sigma_{\Omega_m^G}\right) dt) \end{aligned} \quad (42)$$

α_1, β_1 and ρ represent the *STA* design parameters. The gains must be chosen as (43) in order to ensure the sliding manifolds convergence to zero in finite time (*F.T.C*) [18].

$$\begin{cases} \alpha_1 > \dfrac{\Phi_1}{\Gamma_{m1}} \\ {\beta_1}^2 \ge \dfrac{4\Phi_1 \, \Gamma_{M1}(\alpha_1 + \Phi_1)}{{\Gamma_{m1}}^2 \Gamma_{m1}(\alpha_1 - \Phi_1)} \\ 0 < \rho \le 0.5 \end{cases} \quad (43)$$

The sliding surfaces for the inner d-q stator current components are given as follow:

$$\begin{cases} \sigma_{I_{ds}} = I_{ds}{}^* - I_{ds} \\ \sigma_{I_{qs}} = I_{qs}{}^* - I_{qs} \end{cases} \quad (44)$$

It follows that

$$\begin{cases} \dot{\sigma}_{I_{ds}} = \dot{I}_{ds}^{*} - \dot{I}_{ds} \\ \ddot{\sigma}_{I_{ds}} = \varrho_2(t,x) + \Upsilon_2(t,x)\dot{V}_{ds}^{*} \end{cases} \quad and \quad \begin{cases} \dot{\sigma}_{I_{qs}} = \dot{I}_{qs}^{*} - \dot{I}_{qs} \\ \ddot{\sigma}_{I_{qs}} = \varrho_3(t,x) + \Upsilon_3(t,x)\dot{V}_{qs}^{*} \end{cases} \tag{45}$$

$\varrho_2(t,x), \varrho_3(t,x)$, $\Upsilon_2(t,x)$ and $\Upsilon_3(t,x)$ are the uncertain bounded functions that satisfy

$$\begin{cases} \varrho_2(t,x) > 0, |\varrho_2(t,x)| > \Phi_2, 0 < \Gamma_{m2} < \Upsilon_2 < \Gamma_{M2} \\ \varrho_3(t,x) > 0, |\varrho_3(t,x)| > \Phi_3, 0 < \Gamma_{m3} < \Upsilon_3 < \Gamma_{M3} \end{cases}$$

By designed the controllers using the super twisting algorithm, the control voltages of the *d-q* axis are given as follow:

$$\begin{cases} V_{ds}^{*} = \mu_1 + \mu_2 - Fem_d \\ V_{qs}^{*} = \mathrm{W}_1 + \mathrm{W}_2 + Fem_q \end{cases}$$

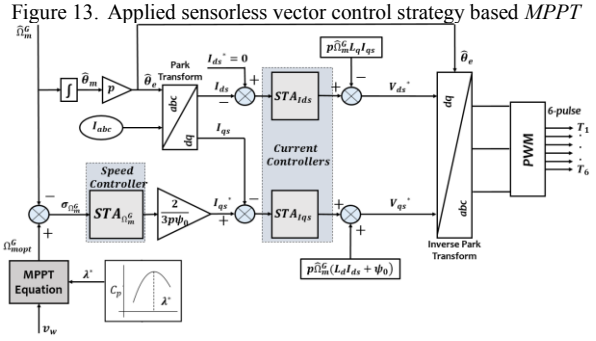
$$Where: \begin{cases} \dot{\mu}_1 = -\alpha_2 sign(\sigma_{I_{ds}}) \\ \mu_2 = -\beta_2 |\sigma_{I_{ds}}|^{\rho} sign(\sigma_{I_{ds}}) \\ Fem_d = p\hat{\Omega}_m^G L_q I_{qs} \end{cases} \tag{46}$$

$$and \quad \begin{cases} \dot{\mathrm{W}}_1 = -\alpha_3 sign(\sigma_{I_{qs}}) \\ \mathrm{W}_2 = -\beta_3 \left|\sigma_{I_{qs}}\right|^{\rho} sign\left(\sigma_{I_{qs}}\right) \\ Fem_q = p\hat{\Omega}_m^G (L_d I_{ds} + \psi_0) \end{cases}$$

Fem_d and Fem_q represent the compensation terms, α_i, β_i and ρ $(i = 2,3)$ are design parameters. To ensure the sliding manifolds convergence to zero in finite time, the gains can be chosen as follow:

$$\begin{cases} \alpha_i > \dfrac{\Phi_i}{\Gamma_{mi}} \\ {\beta_i}^2 \geq \dfrac{4\Phi_i}{{\Gamma_{mi}}^2} \dfrac{\Gamma_{Mi}(\alpha_i + \Phi_i)}{\Gamma_{mi}(\alpha_i - \Phi_i)}, \quad (i = 2,3). \\ 0 < \rho \leq 0.5 \end{cases} \tag{47}$$

Figure 13. Applied sensorless vector control strategy based *MPPT*



V. Results & Discussion

Computer simulation studies of the wind electric water pumping system, the *PI, FLC and SDFLC MRAS* observer are carried out using *Matlab/Simulink* software. The simulation parameters can be found in [4].

- Open loop performance comparison

In the first study, the three adaptation mechanisms were tested and evaluated in open loop mode under random step wind speed profile. The wind speed has been changed from *10m/s* to *3m/s* and finally to *7m/s*.

Figure (14) shows the actual generator rotational speed measured by a real speed sensor and the estimated rotational speed using an *MRAS* observer based on: proportional integral controller, two-input fuzzy logic controller and signed distance fuzzy logic controller adaptation mechanisms. Figure (15) illustrates the estimation errors using the three adaptation mechanisms. From the obtained results, it can be seen that classical *MRAS* observer based on *PI* controller suffers from large estimation error during transient regime and under abruptly wind speed variation. The technique based on two-input fuzzy logic controller has significantly improve the estimation performance of the conventional observer, the design of *49* fuzzy rules makes having an optimal controller a difficult task. Therefore, small oscillations are observed during dynamic regime for this adaptation mechanism. To improve the estimation performance and easily obtain an optimal controller, the signed distance fuzzy logic controller is considered as an adaptation mechanism. Instead of *49* fuzzy rules, only seven fuzzy rules were used. On the same Figures, we can notice high estimation performance in both steady-state and dynamic regime comparing with the conventional *PI* and the *CFLC MRAS* observers.

Figure 14. Comparison of the studied adaptation mechanisms

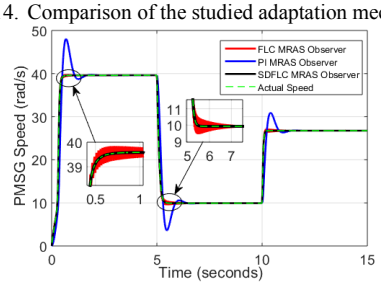
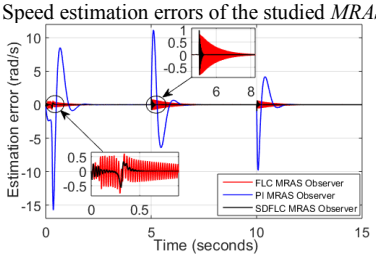


Figure 15. Speed estimation errors of the studied *MRAS* observers



- Closed loop performance

In a second test, the model reference adaptive speed observer based on signed distance fuzzy logic controller adaptation mechanism has been validated in closed loop mode. The *SDFLC MRAS* observer was used to provide the rotational speed information for the vector control strategy shown in Figure (13). Figure (16) shows the applied wind speed profile, the wind speed was modeled as a sum of deterministic several harmonics:

$$v_w(t) = 7 + 0.2\,sin(0.1047t) + 2\,sin(0.2665t) + sin(1.2930t) + 0.2sin(3.6645t) \tag{48}$$

Figures (17) shows the variation of the actual rotational speed, the reference (optimal) speed and the estimated rotational speed using the *SDFLC MRAS* observer. Figure (18) illustrates the speed estimation and the speed tracking errors, for both a very small errors are observed. Figure (19) shows the variation of the turbine mechanical power and the electrical power absorbed by the motor-pump group. A high tracking performance of the maximum mechanical power is obtained by applying the proposed sensorless control. Figure (20) shows the performance of the motor-pump group, its represents the variation of the electrical and the load torque opposed by the centrifugal pump.

Figure 16. Wind speed profile

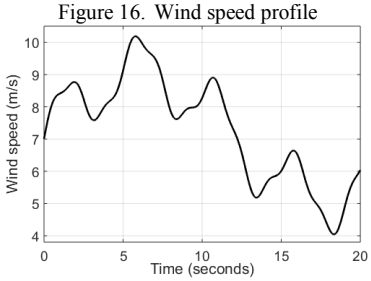


Figure 17. Actual, reference and estimated genarotor rotational speed

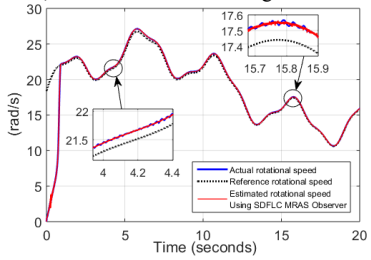


Figure 18. Speed estimation and speed tracking errors

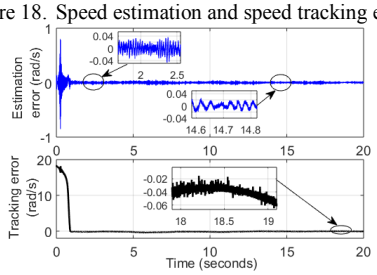


Figure 19. Turbine mechanical power and electrical power absorbed by the motor-pump group

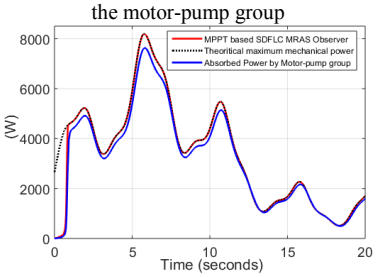
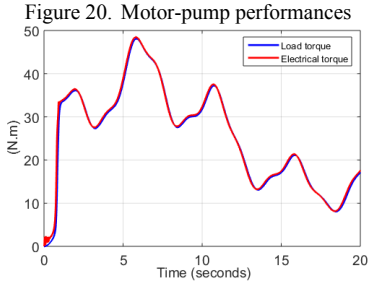


Figure 20. Motor-pump performances



VI. CONCLUSION

In this paper, a sensorless speed control for a wind electric water pumping system based on permanent magnet synchronous generator and permanent magnet *DC* motor has been proposed. A new robust and simple adaptation mechanism based on signed distance fuzzy logic controller has been considered to improve the speed estimation performance of an *MRAS* observer. The obtained results show that the adopted technique provides high performances under different conditions. The *SDFLC* has proven to be an attractive adaptation mechanism in an *MRAS* observer comparing with the others techniques.

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